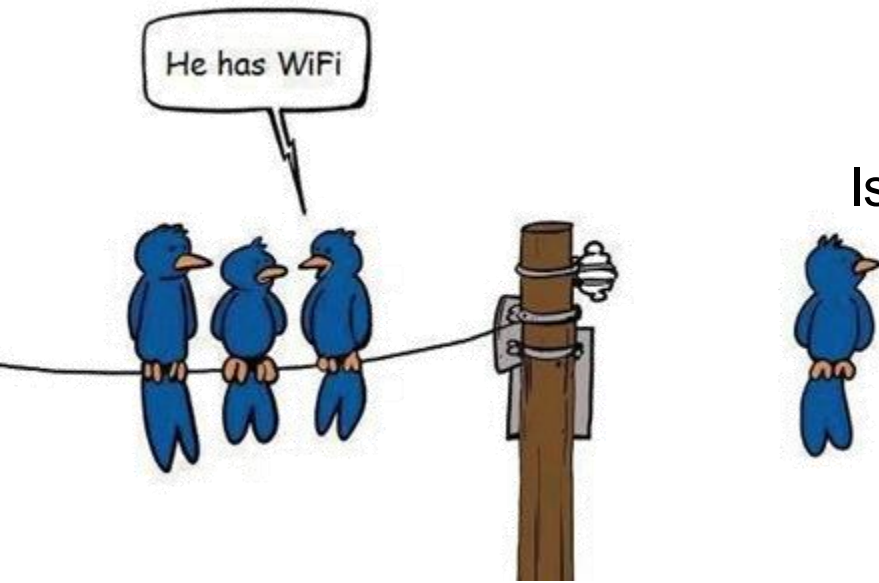


ECSE 4560/ 6560

Modern Communication Systems

Lecture 17: MIMO Diversity

Ish Jain, Fall 2025



Announcements

- Join Piazza: <https://piazza.com/rpi/fall2025/ecse45606560>
- Per-Class Quiz
 - Posted on LMS; Due Today 11:59 PM
- Attendance Link: <https://forms.office.com/r/p6P9YrN1d8>
 - Today's Magic Word: **Diversity**
- Projects
 - Group formation [MCS_Fall25_project.xlsx](#)
 - Project mid-term report due **Friday Nov 7**
 - Final Project Presentation and Demos: Dec 9 and Dec 12
 - Project Report and peer evaluation: Finals exam week
- HW/Exams Dates
 - HW5 coming soon
 - Exam 1: Friday Oct 24 Graded (Submit regrading request by Tue **11/4**)
 - Exam 2: **Friday Nov 21**

Extra Reading on Piazza

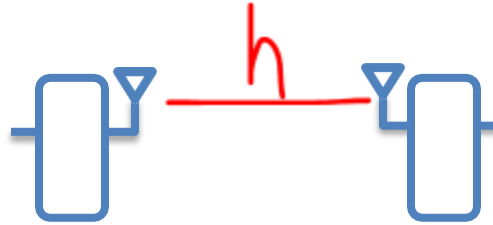
- Principles of Modern Wireless Communication Systems Theory and Practice; by Aditya K. Jagannatham, IIT Kanpur Book
 - https://piazza.com/class_profile/get_resource/mevvk9kirhg2fu/mgb9esiush64ae
 - Parts of Chapter 1,2,3,4,7
 - **MIMO Chapter 6**

Today's Class

- MIMO Gains
 - Receiver Diversity
 - Transmitter Diversity
 - Space-Time codes (Alamouti Codes)
- MIMO Multiplexing Gains

MIMO: Multiple Input Multiple Output

So far: single input single output



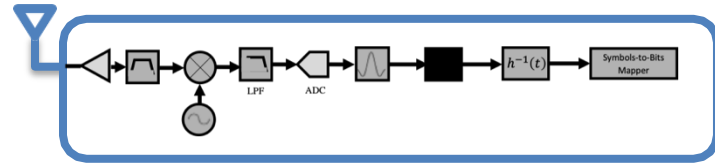
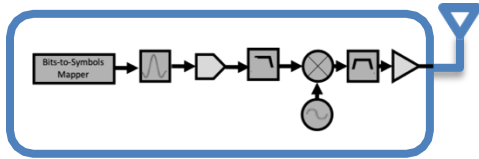
MIMO: multiple input multiple output



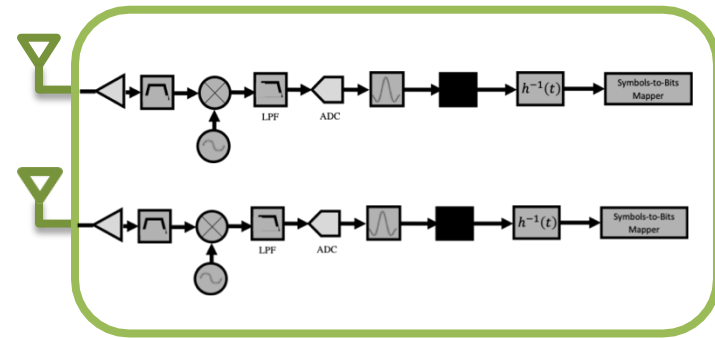
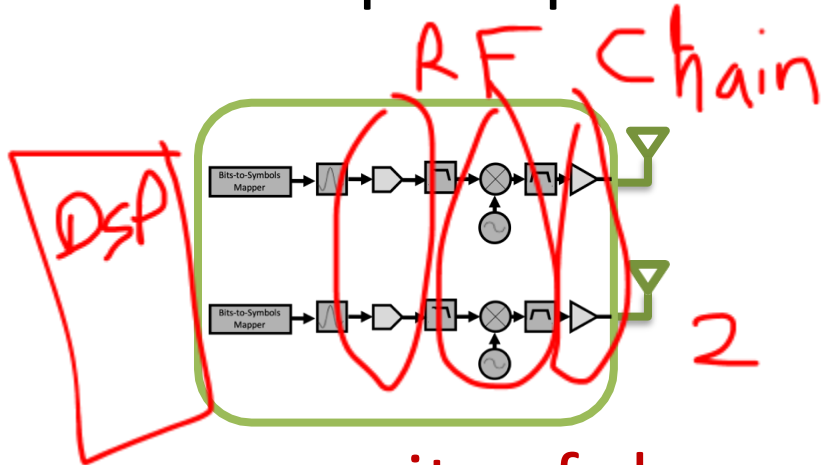
Increase capacity of channel using multiple transmit
and receive antennas.

MIMO: Multiple Input Multiple Output

So far: single input single output

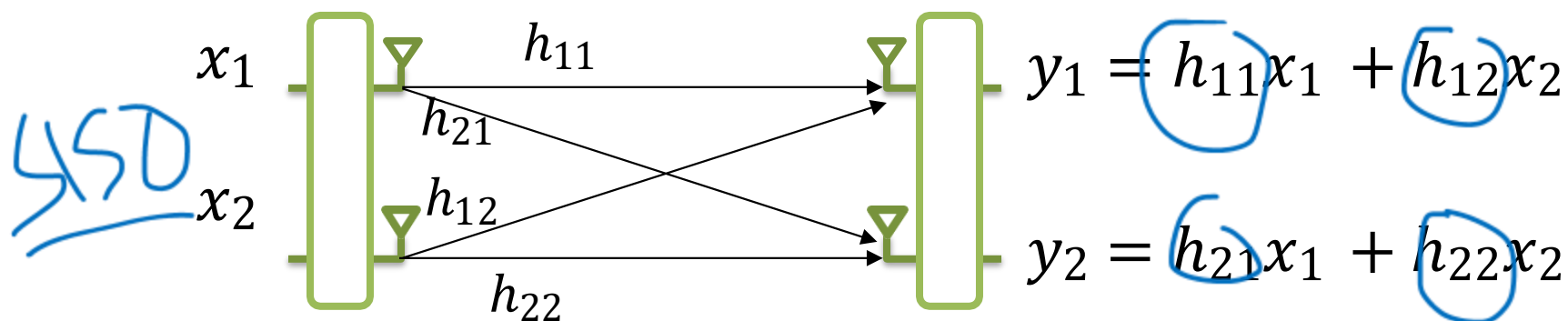


MIMO: multiple input multiple output

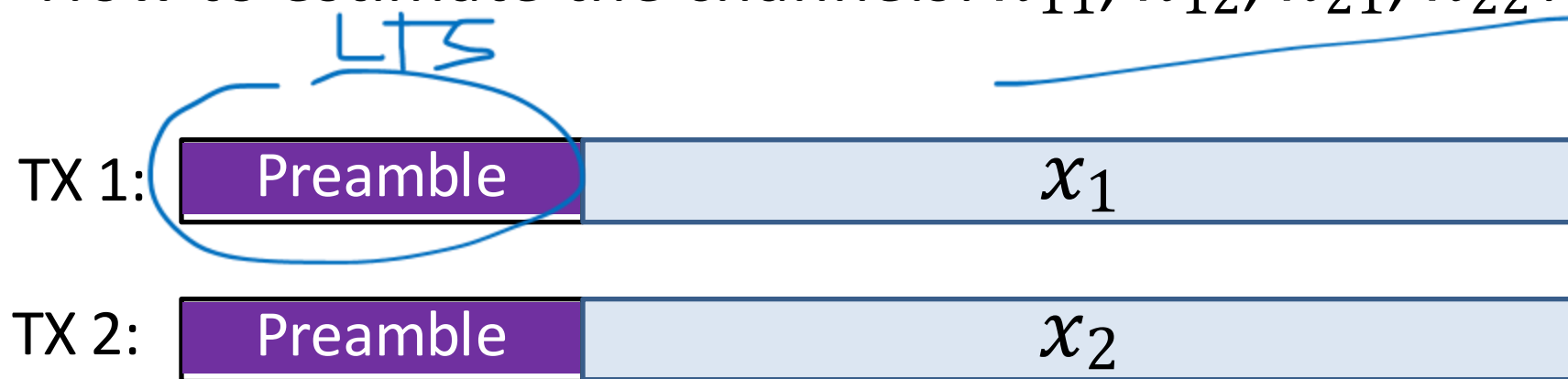


Increase capacity of channel using multiple transmit and receive antennas.

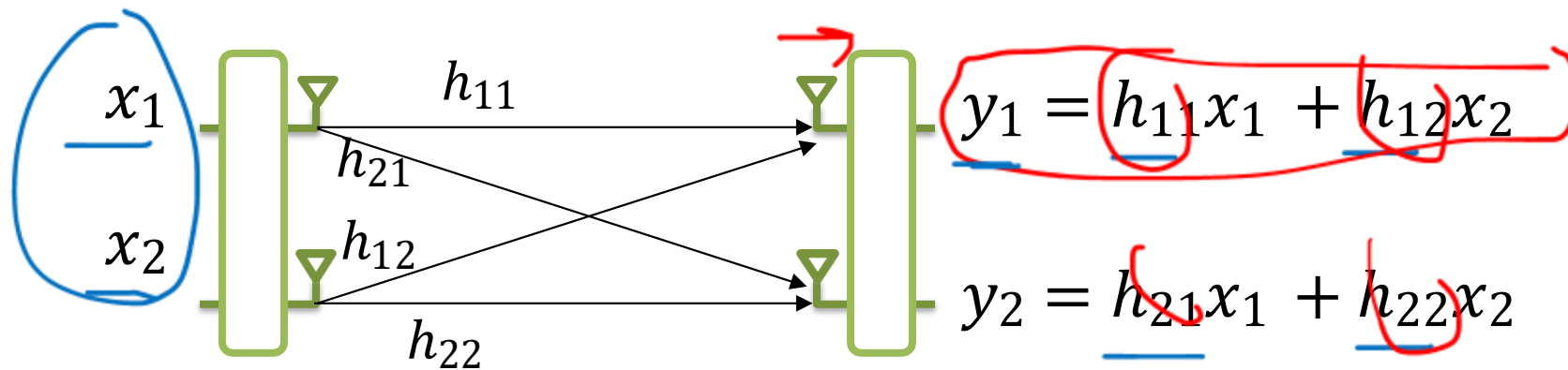
MIMO Channel



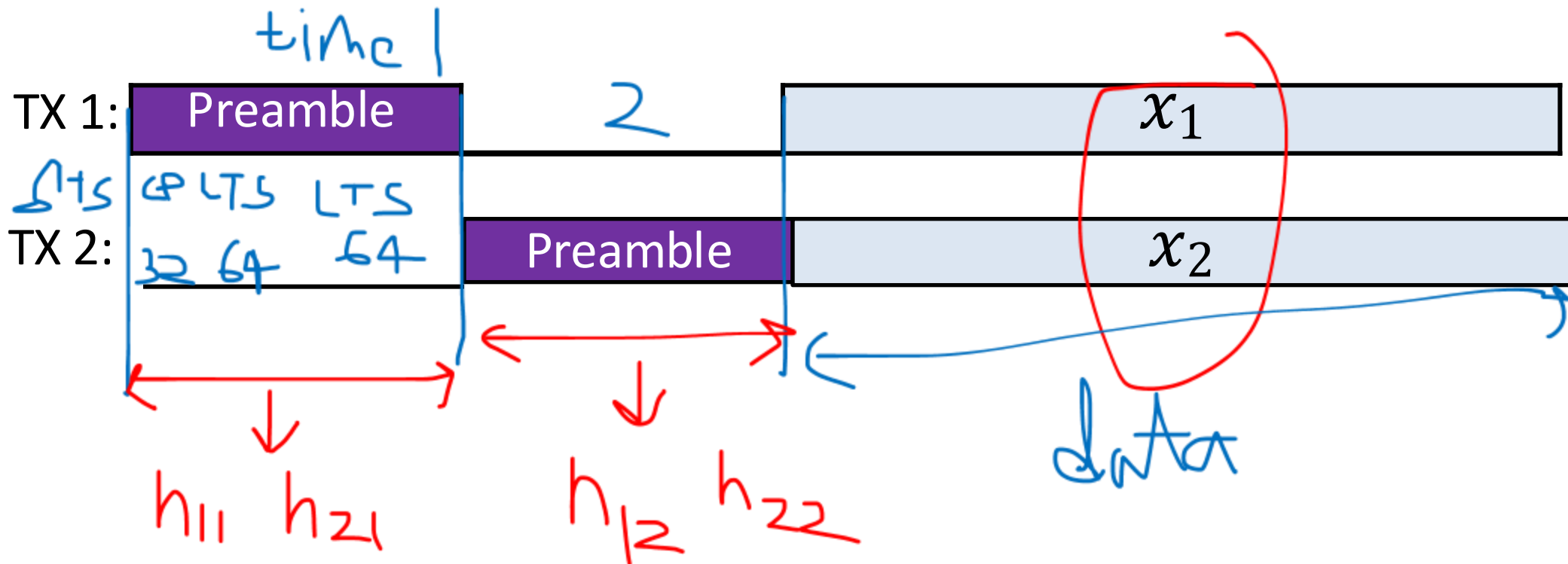
How to estimate the channels: $h_{11}, h_{12}, h_{21}, h_{22}$?



MIMO Channel



How to estimate the channels: h_{11} , h_{12} , h_{21} , h_{22} ?



Today's Class

- MIMO Intro
 - MIMO in WiFi Standards
 - MIMO Intro
 - MIMO antenna-space vector representation
 - MIMO Channel
 - Channel representation
 - Preamble based channel estimation
 - **MIMO Gains**
 - **Receiver Diversity**
 - **Transmitter Diversity**
- 

MIMO Gains

Multiplexing Gain:

- Send multiple packets at the same time
- $N \times N$ MIMO $\rightarrow$ $N \times$ more packets

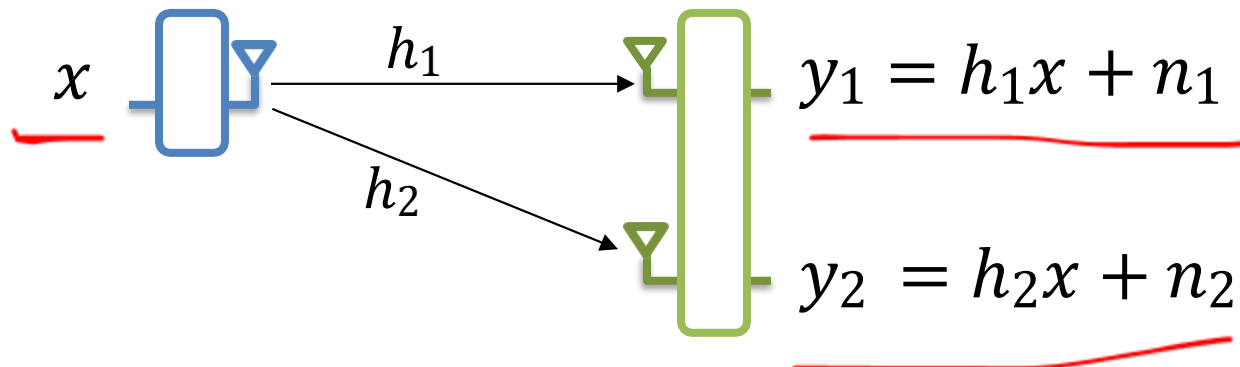
Diversity Gain:

- Send/Receive the same packet on multiple antennas
- Increase SNR of the received packets
 $\rightarrow$ transmit at higher data rates

SNR $\uparrow$ MCS $\uparrow$

Receiver Diversity

51 M0



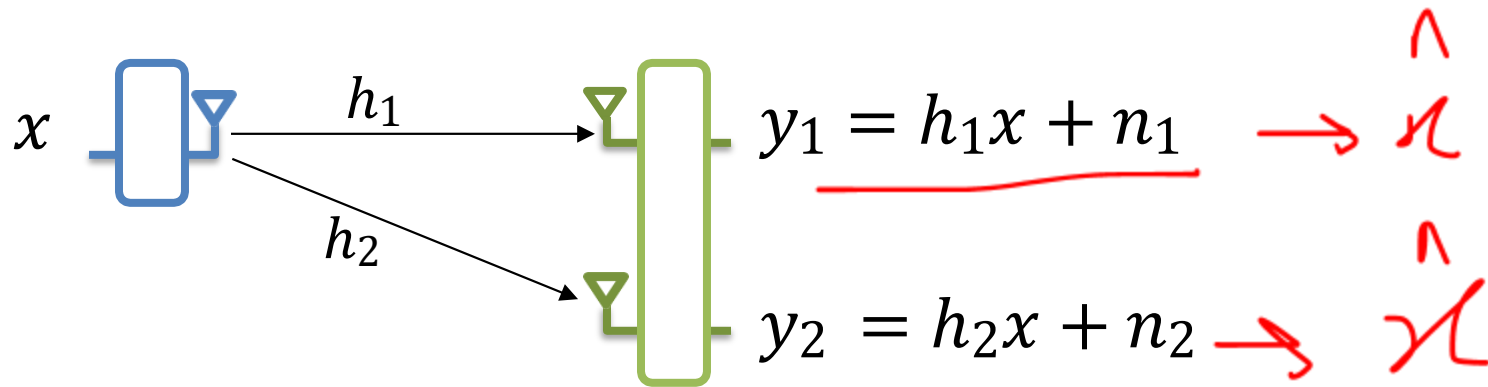
How to best decode x ?

Option 1: Add the received signals

$$\begin{aligned} y_1 + y_2 &= h_1x + n_1 + h_2x + n_2 \\ \underline{\quad} \quad \underline{\quad} \quad \underline{\quad} \\ 2 &= (h_1 + h_2)x + n_1 + n_2 \\ \quad \underline{\quad} \quad \underline{\quad} \end{aligned}$$

Channels can sum up destructively! $h_1 + h_2 \approx 0$

Receiver Diversity



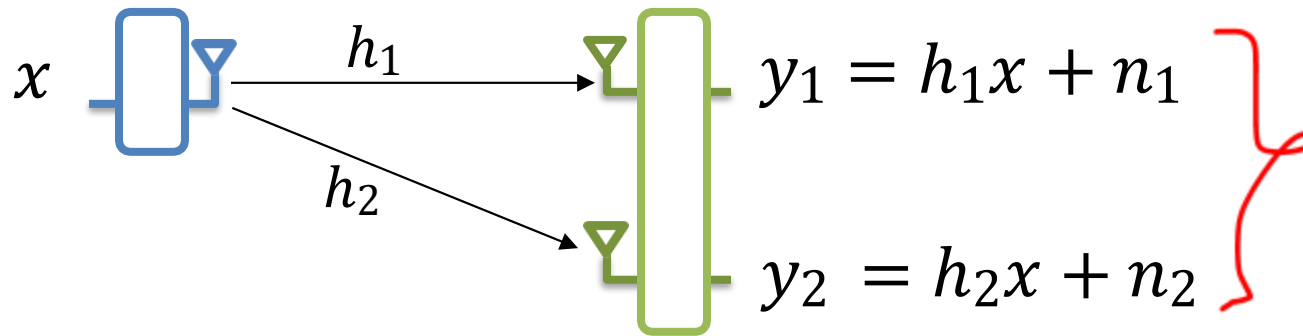
How to best decode x ?

Option 1: Add the received signals

Option 2: ~~Decode independently~~

Sub-optimal!

Receiver Diversity



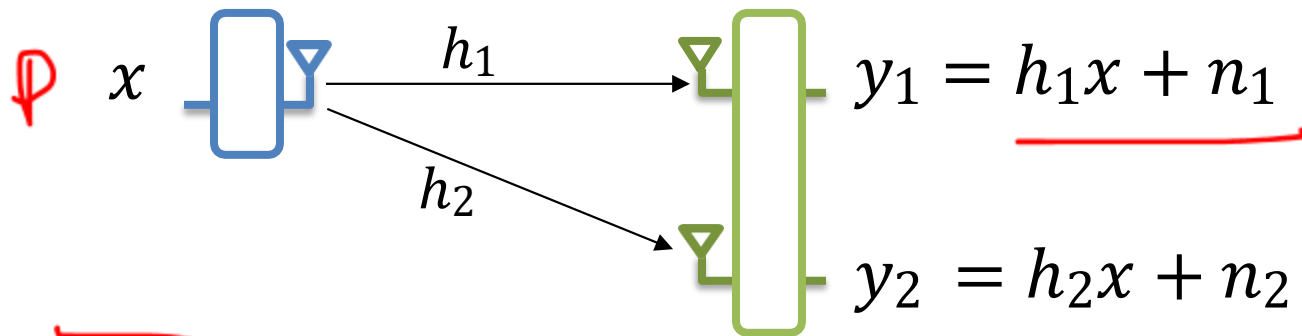
How to best decode x ?

Option 1: Add the received signals

Option 2: Decode independently

Optimal Solution: Maximum Ratio Combining (MRC)

Maximum Ratio Combining

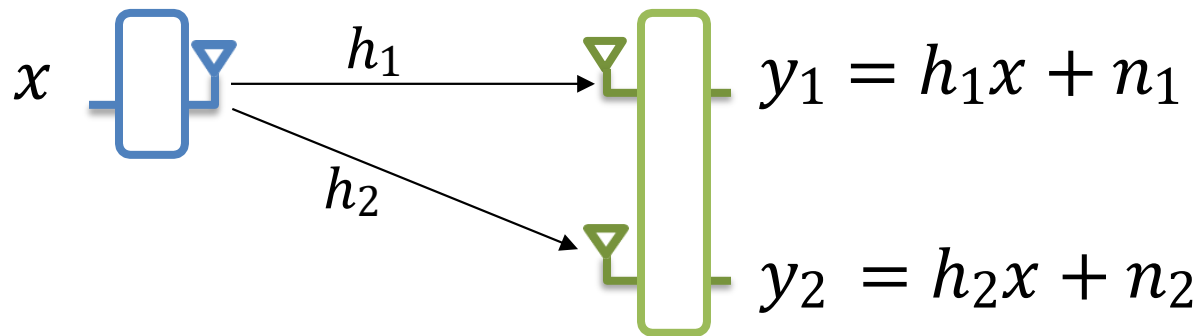


$$\begin{aligned} \left| \alpha_1 y_1 + \alpha_2 y_2 \right|^2 &= h_1^* y_1 + h_2^* y_2 = \underbrace{h_1^* h_1 x + h_1^* n_1}_{\text{}} + \underbrace{h_2^* h_2 x + h_2^* n_2}_{\text{}} \\ &= \underbrace{(|h_1|^2 + |h_2|^2)}_{\text{}} x + \underbrace{h_1^* n_1 + h_2^* n_2}_{\text{}} \end{aligned}$$

Let $P = E[|x|^2]$ and $\sigma^2 = E[|n_1|^2] = E[|n_2|^2]$

$$\begin{aligned} \text{Signal Power} &= E[|(|h_1|^2 + |h_2|^2)x|^2] \\ &= (|h_1|^2 + |h_2|^2)^2 E[|x|^2] \\ &= (|h_1|^2 + |h_2|^2)^2 P \end{aligned}$$

Maximum Ratio Combining



$$\begin{aligned}\alpha_1 y_1 + \alpha_2 y_2 &= h_1^* y_1 + h_2^* y_2 = h_1^* h_1 x + h_1^* n_1 + h_2^* h_2 x + h_2^* n_2 \\ &= (|h_1|^2 + |h_2|^2)x + \underline{h_1^* n_1 + h_2^* n_2}\end{aligned}$$

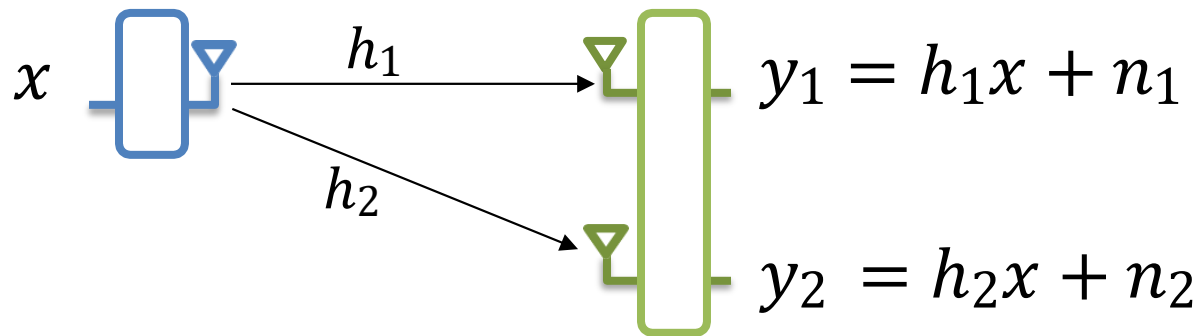
Let $P = E[|x|^2]$ and $\sigma^2 = E[|n_1|^2] = E[|n_2|^2]$

$$\text{Signal Power} = \underline{(|h_1|^2 + |h_2|^2)^2 P}$$

$$\underline{\text{Noise Power}} = E[|h_1^* n_1 + h_2^* n_2|^2] = E[|h_1^* n_1|^2] + E[|h_2^* n_2|^2]$$

$$= \underline{|h_1|^2 E[|n_1|^2]} + \underline{|h_2|^2 E[|n_2|^2]} = \underline{(|h_1|^2 + |h_2|^2) \sigma^2}$$

Maximum Ratio Combining



$$\begin{aligned}\alpha_1 y_1 + \alpha_2 y_2 &= h_1^* y_1 + h_2^* y_2 = h_1^* h_1 x + h_1^* n_1 + h_2^* h_2 x + h_2^* n_2 \\ &= (|h_1|^2 + |h_2|^2)x + h_1^* n_1 + h_2^* n_2\end{aligned}$$

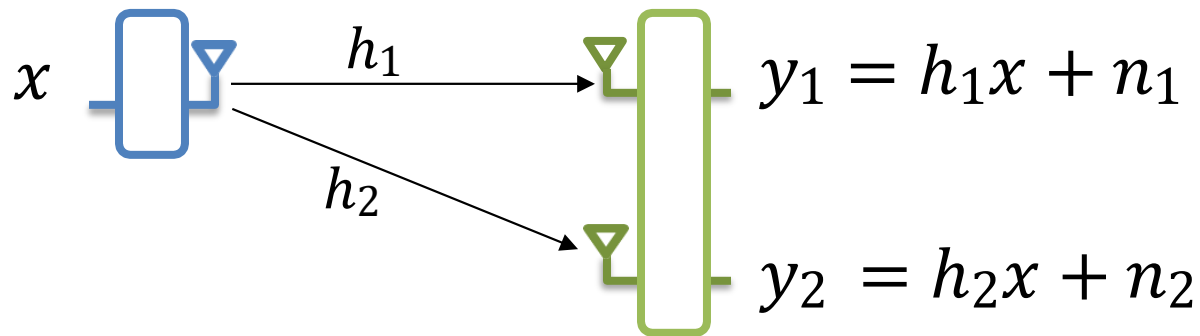
Let $P = E[|x|^2]$ and $\sigma^2 = E[|n_1|^2] = E[|n_2|^2]$

$$\text{Signal Power} = (|h_1|^2 + |h_2|^2)^2 P$$

$$\text{Noise Power} = (|h_1|^2 + |h_2|^2)\sigma^2$$

$$SNR = \frac{(|h_1|^2 + |h_2|^2)^2 P}{(|h_1|^2 + |h_2|^2)\sigma^2} = \underbrace{(|h_1|^2 + |h_2|^2)}_{\text{MRC Gain}} \boxed{\frac{P}{\sigma^2}}$$

Maximum Ratio Combining



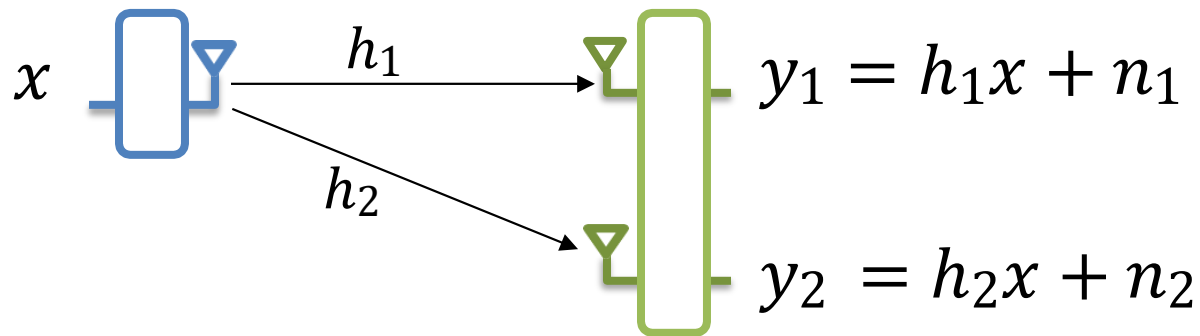
$$\begin{aligned}\alpha_1 y_1 + \alpha_2 y_2 &= h_1^* y_1 + h_2^* y_2 = h_1^* h_1 x + h_1^* n_1 + h_2^* h_2 x + h_2^* n_2 \\ &= (|h_1|^2 + |h_2|^2)x + h_1^* n_1 + h_2^* n_2\end{aligned}$$

With Receiver Diversity: $SNR = (|h_1|^2 + |h_2|^2) \frac{P}{\sigma^2}$

MRC

Single Receiver: $SNR = |h_1|^2 \frac{P}{\sigma^2}$

Receiver Diversity Gain



With Receiver Diversity: $SNR = (|h_1|^2 + |h_2|^2) \frac{P}{\sigma^2} = 2 |h_1|^2 \frac{P}{\sigma^2}$

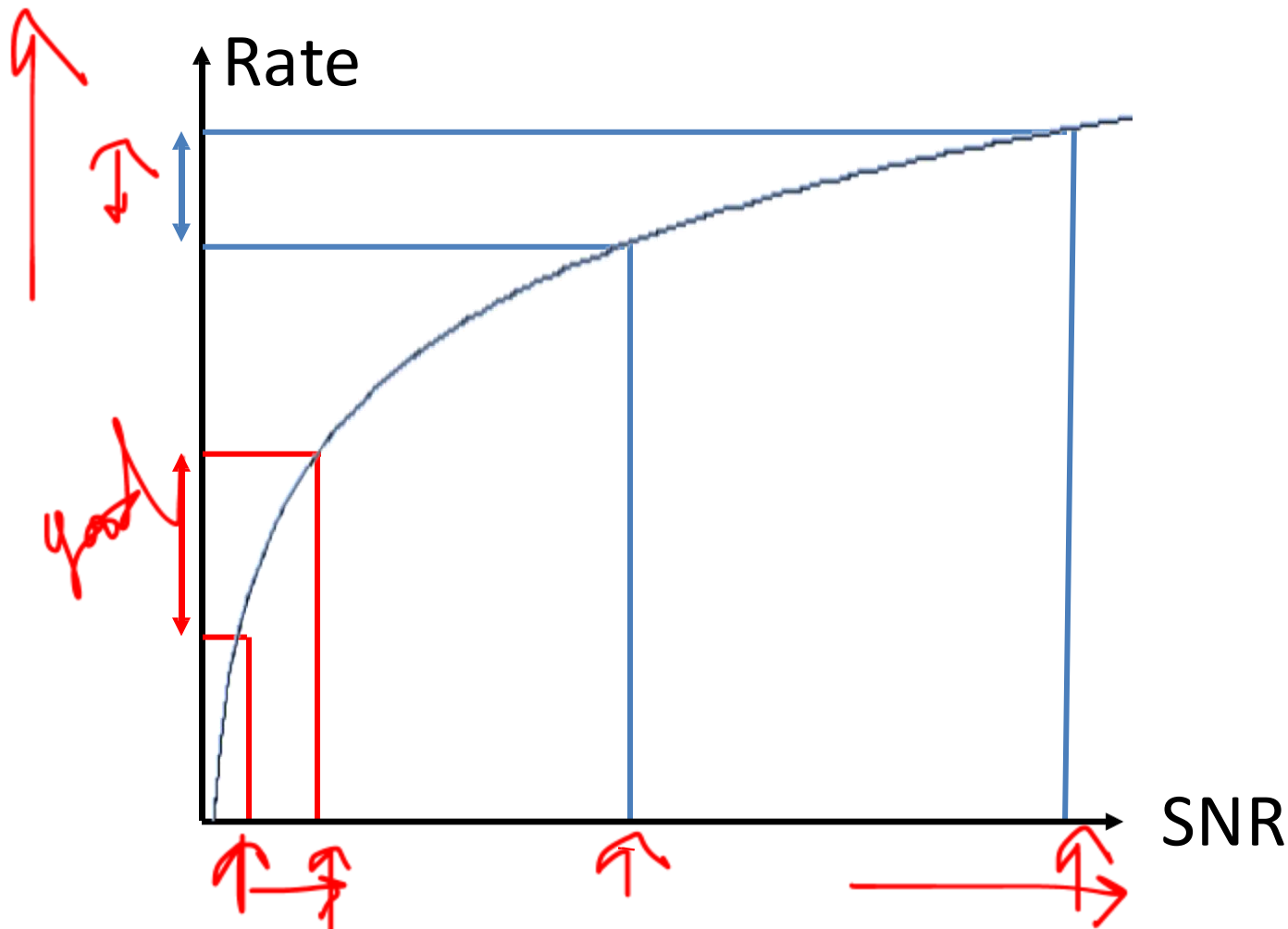
Single Receiver: $SNR = |h_1|^2 \frac{P}{\sigma^2}$

- $|h_1|^2 \approx |h_2|^2$ $\rightarrow$ Can double SNR!

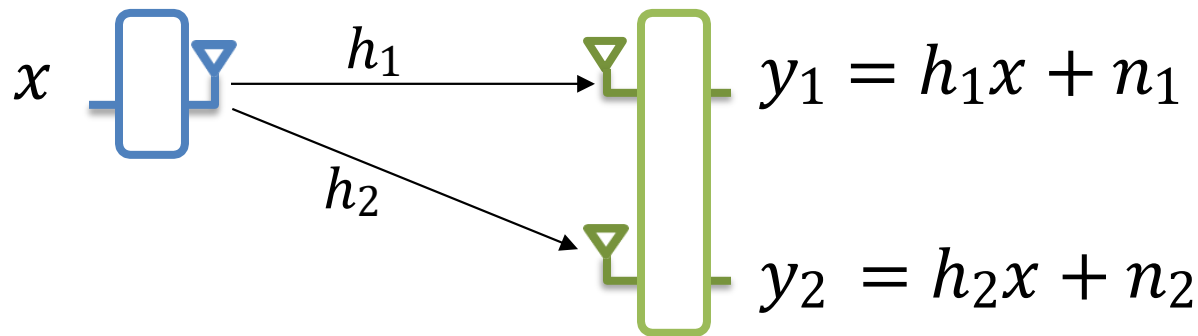
Receiver Diversity Gain

Do we care about doubling the SNR?

$$Capacity \propto \log(SNR)$$



Receiver Diversity Gain



With Receiver Diversity: $SNR_{2R)} = (|h_1|^2 + |h_2|^2) \frac{P}{\sigma^2}$

Single Receiver: $SNR_{1R)} = |h_1|^2 \frac{P}{\sigma^2}$

• $|h_1|^2 \approx |h_2|^2 \rightarrow$ Can double SNR!

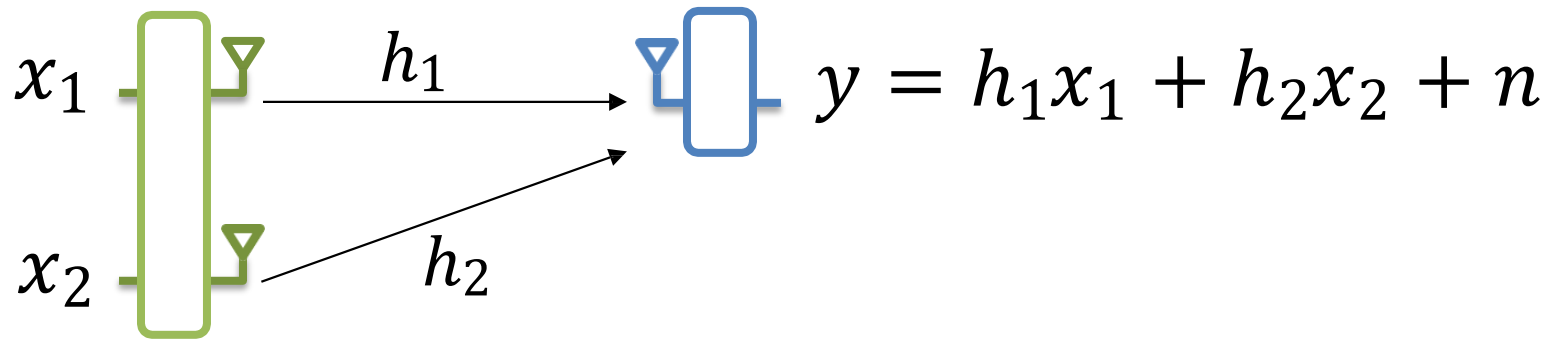
• $|h_1|^2 \ll |h_2|^2 \rightarrow$ Huge Gain in SNR

• $|h_1|^2 \gg |h_2|^2 \rightarrow$ Little Gain in SNR

Today's Class

- MIMO Channel
 - Channel representation
 - Preamble based channel estimation
- MIMO Gains
 - Receiver Diversity
 - **Transmitter Diversity**
 - Space-Time codes (Alamouti Codes)
- MIMO Multiplexing Gains

Transmitter Diversity



What should we transmit on each antenna?

Option 1: transmit the same thing x on both antennas

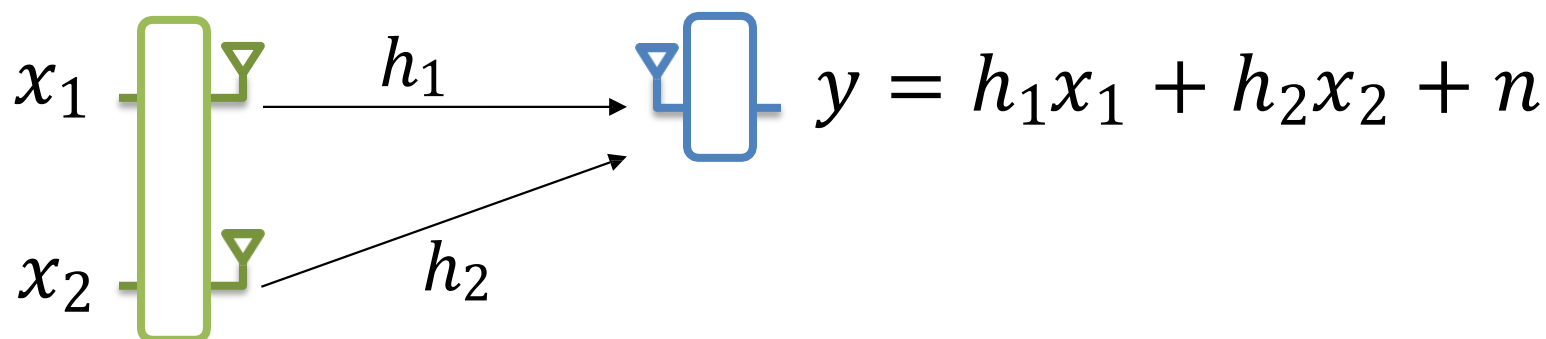
$$y = (h_1 + h_2)x + n$$

Channels can sum up destructively! $h_1 + h_2 \approx 0$

Total transmit power = $2P \rightarrow$ Doubled TX power

$\rightarrow$ Why not use 1 TX with $2P$

Transmitter Diversity

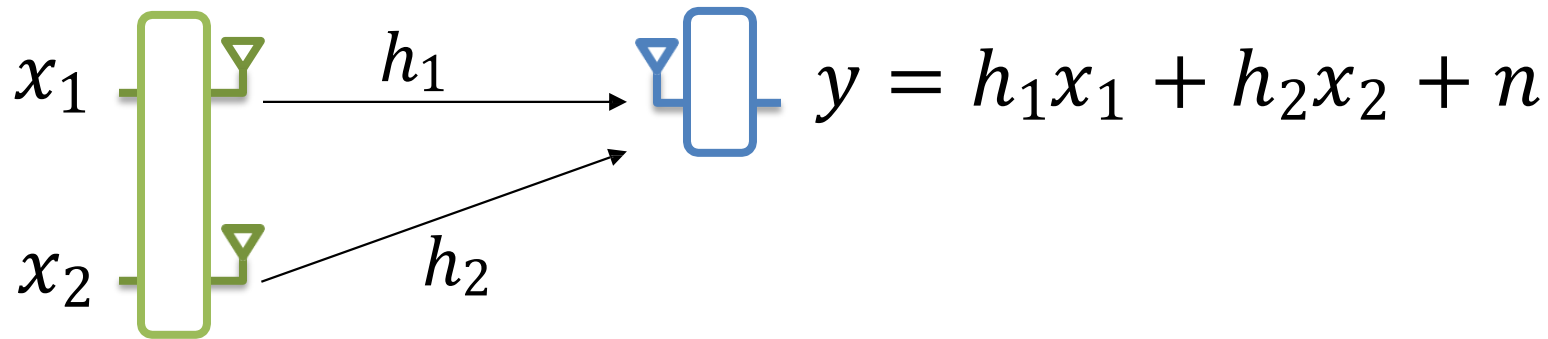


What should we transmit on each antenna?

Option 1: transmit the same thing x on both antennas

- Must ensure signals sum up constructively.
- Must ensure total TX power = $E[|x_1|^2] + E[|x_2|^2]$
 $= E[|x|^2] = P$

Transmitter Diversity



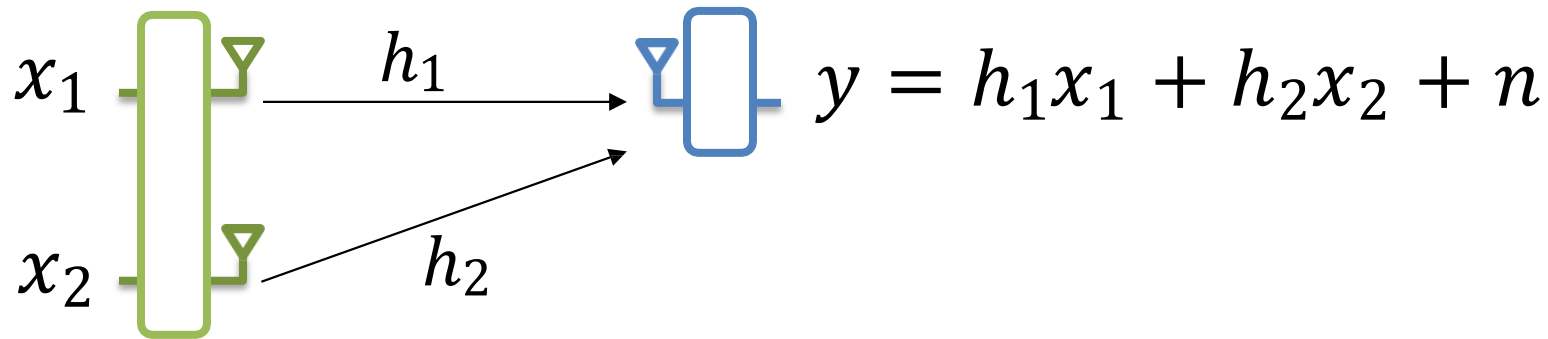
What should we transmit on each antenna?

Option 2: Maximum Ratio Combining (MRC)

$$\left. \begin{array}{l} x_1 = \alpha_1 x \\ x_2 = \alpha_2 x \end{array} \right\} y = (\alpha_1 h_1 + \alpha_2 h_2)x + n$$

$$\begin{aligned} \text{Set: } \alpha_1 = h_1^*, \alpha_2 = h_2^* &\rightarrow y = (h_1^* h_1 + h_2^* h_2)x + n \\ &= (|h_1|^2 + |h_2|^2)x + n \end{aligned}$$

Transmitter Diversity



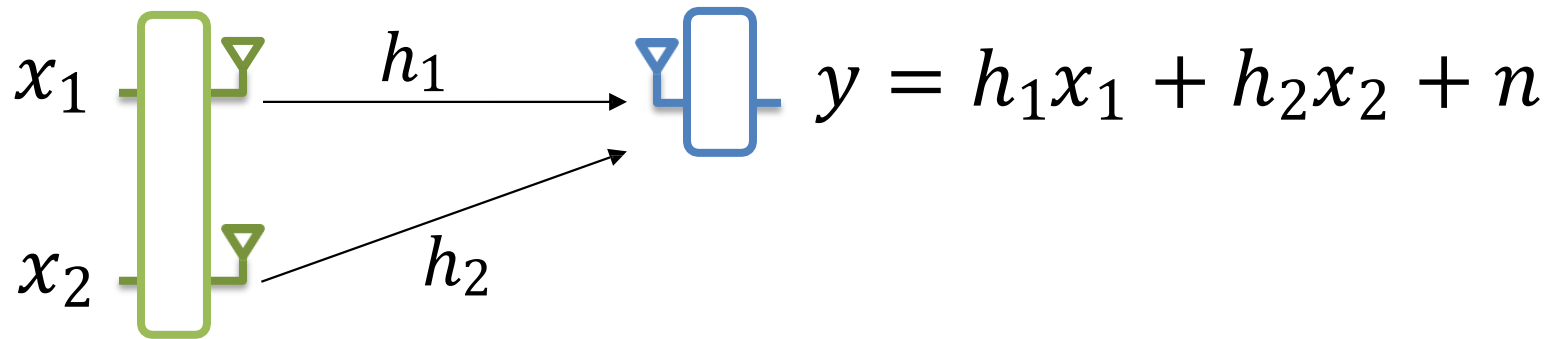
What should we transmit on each antenna?

Option 2: Maximum Ratio Combining (MRC)

$$x_1 = \frac{h_1^*}{\sqrt{|h_1|^2 + |h_2|^2}} x \quad x_2 = \frac{h_2^*}{\sqrt{|h_1|^2 + |h_2|^2}} x$$

$$\begin{aligned} \text{Total TX Power} &= \mathbb{E}[|x_1|^2] + \mathbb{E}[|x_2|^2] \\ &= \frac{|h_1|^2}{|h_1|^2 + |h_2|^2} P + \frac{|h_2|^2}{|h_1|^2 + |h_2|^2} P = P \end{aligned}$$

Transmitter Diversity



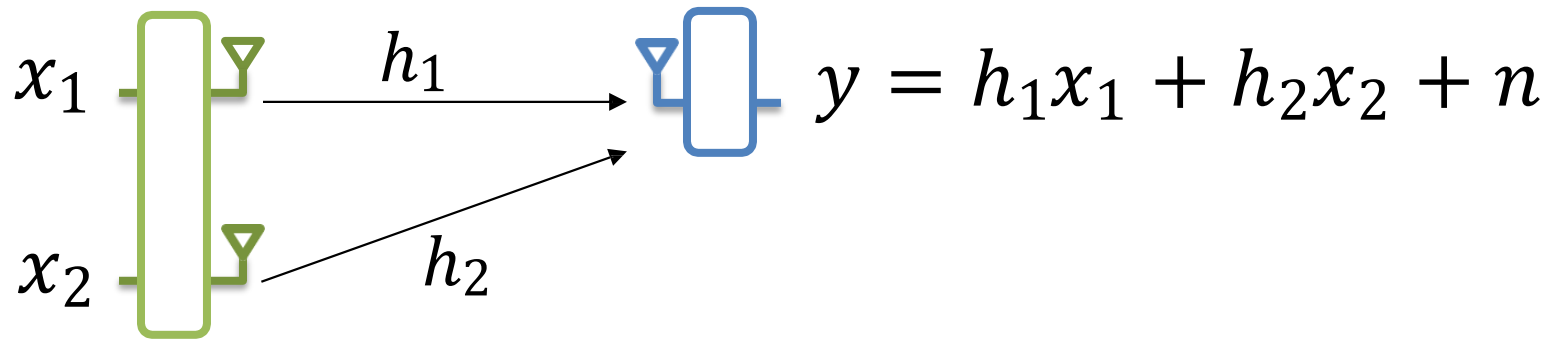
What should we transmit on each antenna?

Option 2: Maximum Ratio Combining (MRC)

$$x_1 = \frac{h_1^*}{\sqrt{|h_1|^2 + |h_2|^2}} x \quad x_2 = \frac{h_2^*}{\sqrt{|h_1|^2 + |h_2|^2}} x$$

$$y = \frac{h_1 h_1^*}{\sqrt{|h_1|^2 + |h_2|^2}} x + \frac{h_2 h_2^*}{\sqrt{|h_1|^2 + |h_2|^2}} x + n$$

Transmitter Diversity



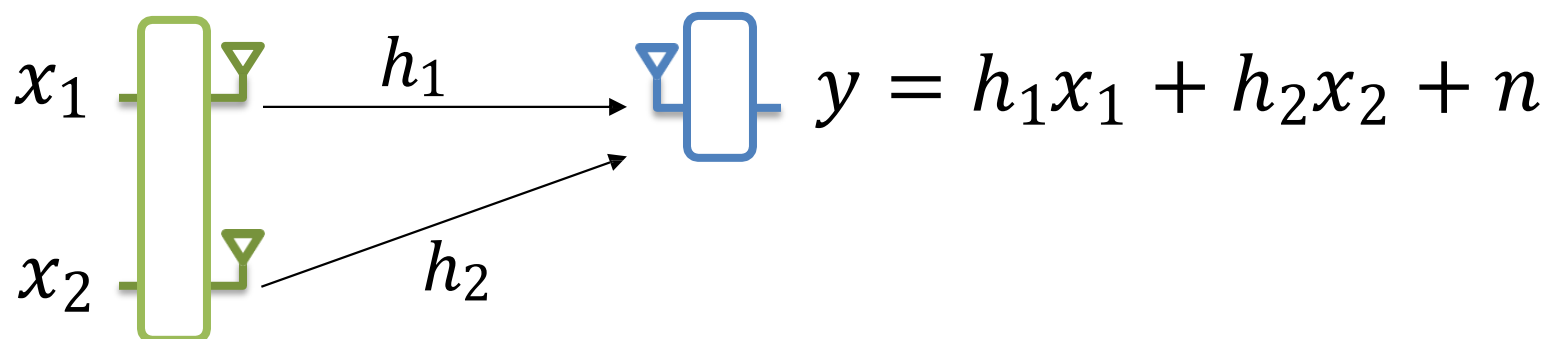
What should we transmit on each antenna?

Option 2: Maximum Ratio Combining (MRC)

$$x_1 = \frac{h_1^*}{\sqrt{|h_1|^2 + |h_2|^2}} x \quad x_2 = \frac{h_2^*}{\sqrt{|h_1|^2 + |h_2|^2}} x$$

$$y = \frac{|h_1|^2 + |h_2|^2}{\sqrt{|h_1|^2 + |h_2|^2}} x + n = \left(\sqrt{|h_1|^2 + |h_2|^2} \right) x + n$$

Transmitter Diversity



What should we transmit on each antenna?

Option 2: Maximum Ratio Combining (MRC)

$$y = \left(\sqrt{|h_1|^2 + |h_2|^2} \right) x + n$$

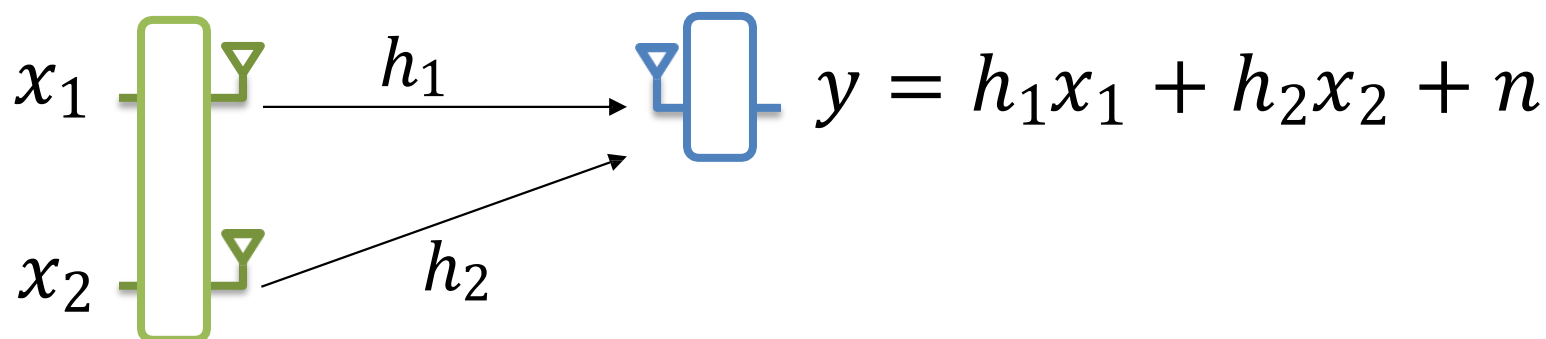
$$SNR = (|h_1|^2 + |h_2|^2) \frac{P}{\sigma^2} \rightarrow \text{Similar SNR Gain to RX Diversity}$$

Caveat: MRC at TX Requires Feedback from the Receiver!

Today's Class

- MIMO Channel
 - Channel representation
 - Preamble based channel estimation
- MIMO Gains
 - Receiver Diversity
 - Transmitter Diversity
 - **Space-Time codes (Alamouti Codes)**
- MIMO Multiplexing Gains

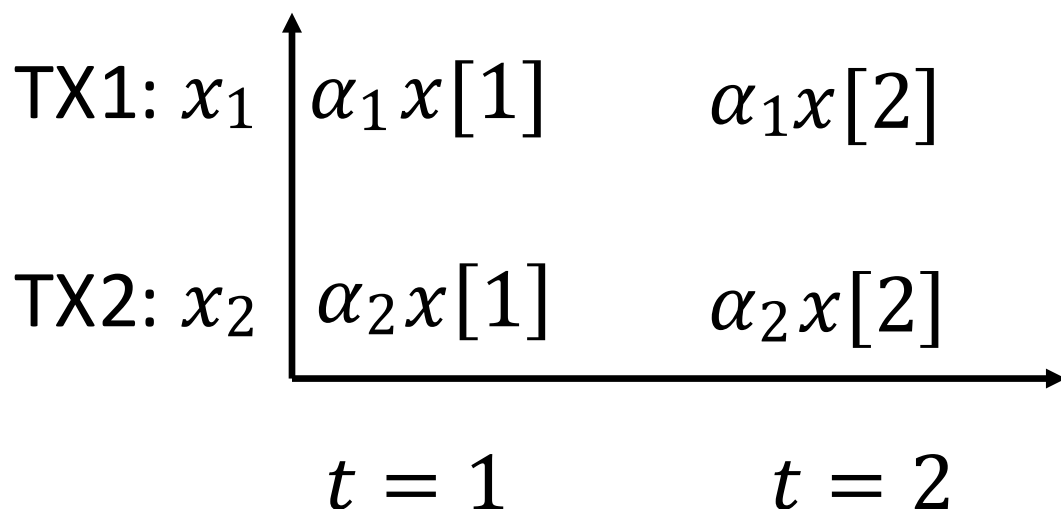
Transmitter Diversity



What should we transmit on each antenna?

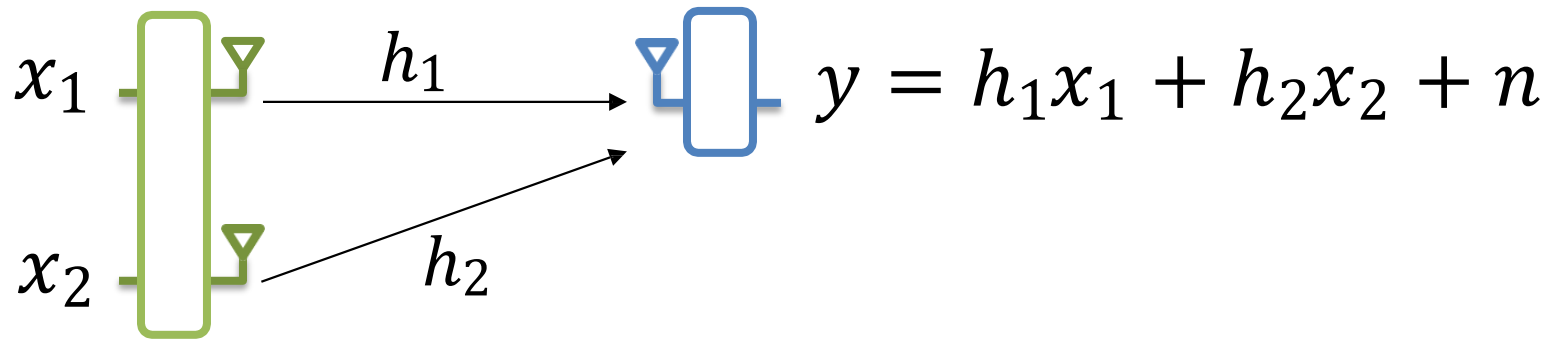
Solution: Use Space-Time Codes

MRC codes across space only (Requires Channel Feedback):



$$\alpha_1 = \frac{h_1^*}{\sqrt{|h_1|^2 + |h_2|^2}}$$
$$\alpha_2 = \frac{h_2^*}{\sqrt{|h_1|^2 + |h_2|^2}}$$

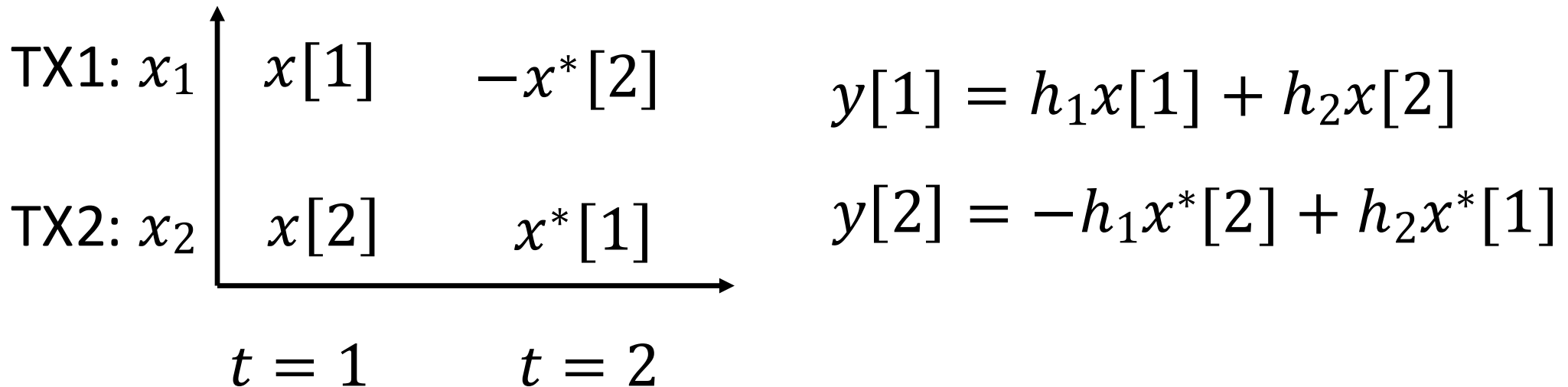
Transmitter Diversity



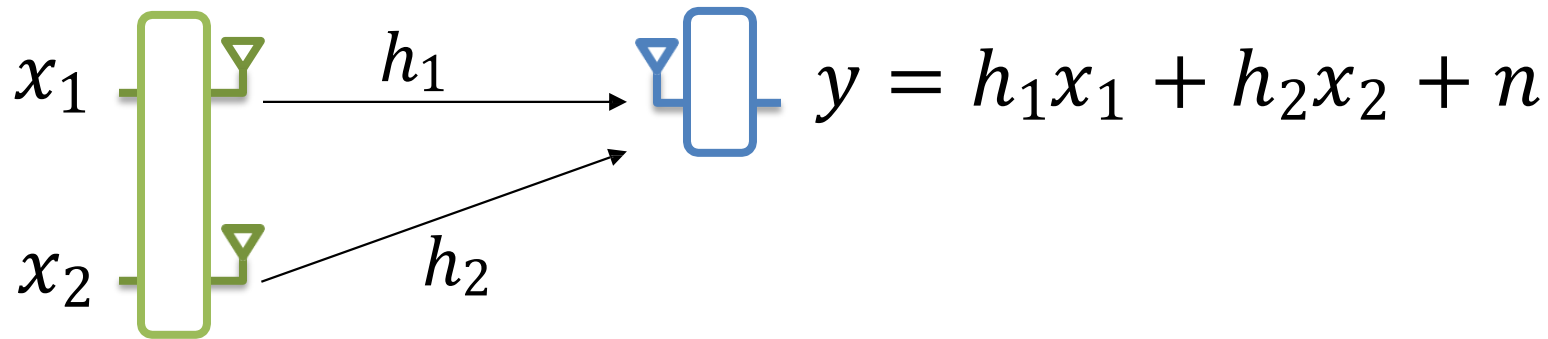
What should we transmit on each antenna?

Solution: Use Space-Time Codes

Alamouti Codes:



Transmitter Diversity



What should we transmit on each antenna?

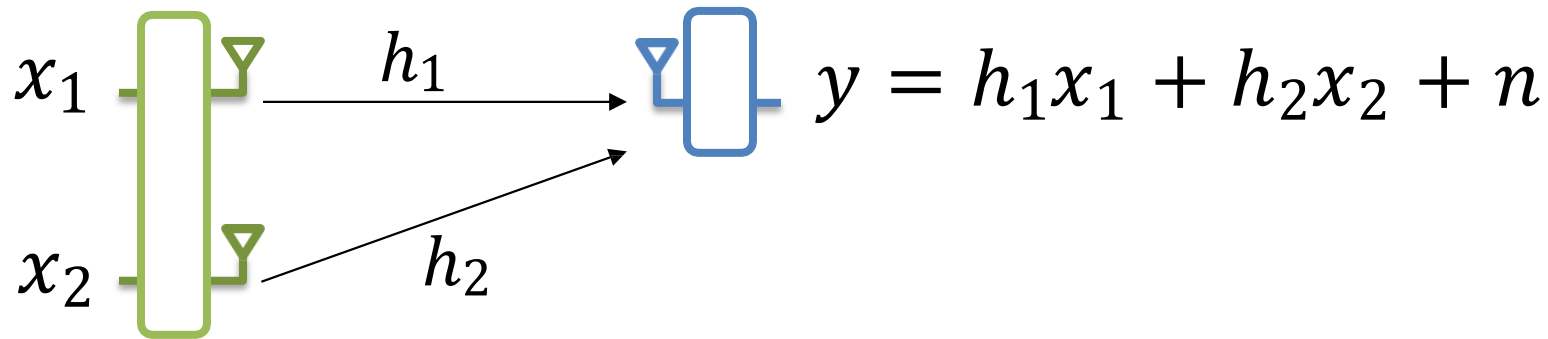
Solution: Use Space-Time Codes

Alamouti Codes:

$$y[1] = h_1x[1] + h_2x[2] \quad y[2] = -h_1x^*[2] + h_2x^*[1]$$

$$\begin{aligned} h_1^*y[1] + h_2y^*[2] &= h_1^*h_1x[1] + h_1^*h_2x[2] \\ &\quad - h_2h_1^*x[2] + h_2h_2^*x[1] \end{aligned}$$

Transmitter Diversity



What should we transmit on each antenna?

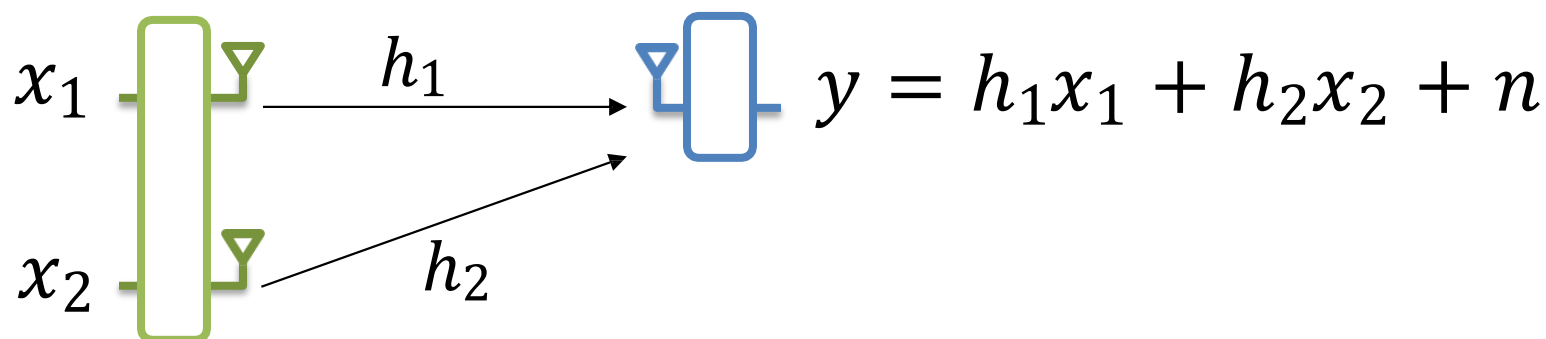
Solution: Use Space-Time Codes

Alamouti Codes:

$$y[1] = h_1x[1] + h_2x[2] \quad y[2] = -h_1x^*[2] + h_2x^*[1]$$

$$\begin{aligned} h_2^*y[1] - h_1y^*[2] &= \cancel{h_2^*h_1x[1]} + h_2^*h_2x[2] \\ &\quad + h_1h_1^*x[2] - \cancel{h_1h_2^*x[1]} \\ &= (|h_1|^2 + |h_2|^2)x[2] \end{aligned}$$

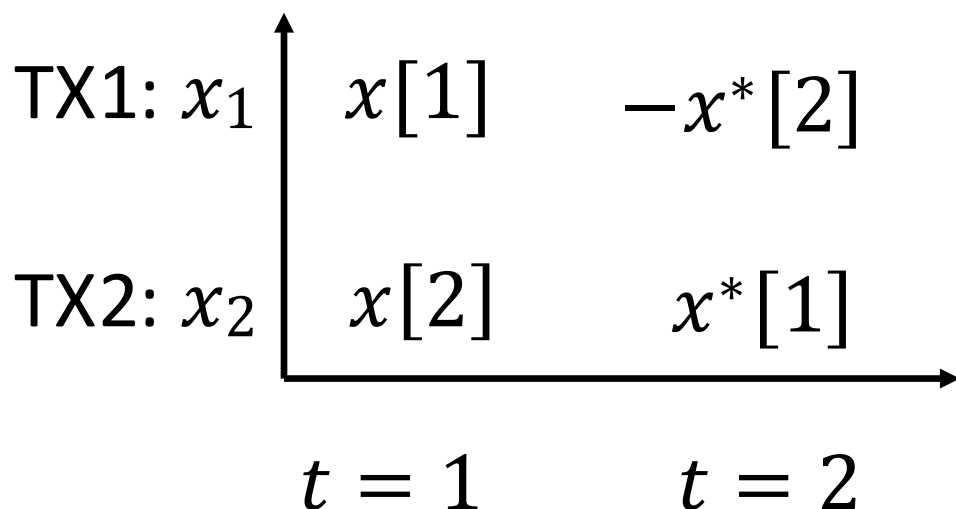
Transmitter Diversity



What should we transmit on each antenna?

Solution: Use Space-Time Codes

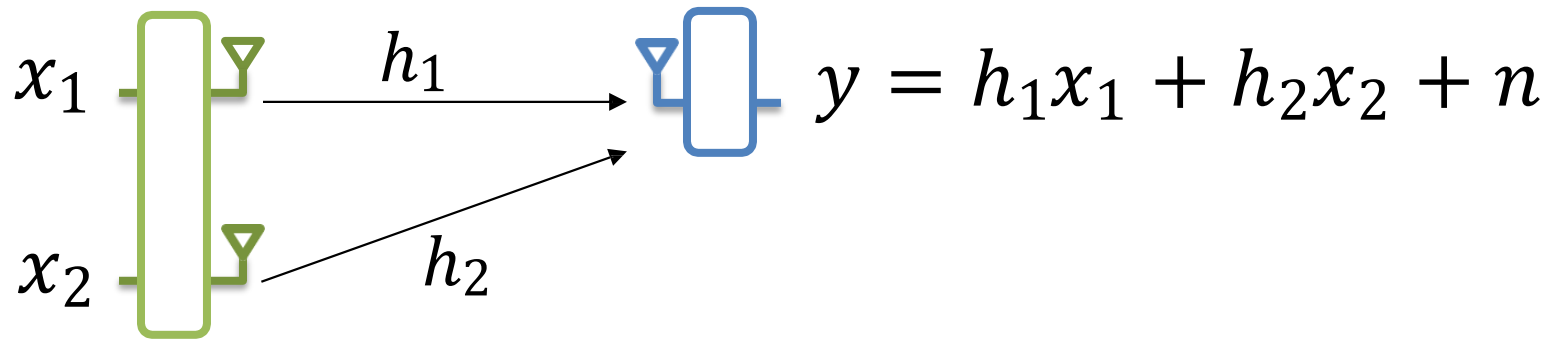
Alamouti Codes:



$$h_1^* y[1] + h_2 y^*[2] = (|h_1|^2 + |h_2|^2) x[1]$$

$$h_2^* y[1] - h_1 y^*[2] = (|h_1|^2 + |h_2|^2) x[2]$$

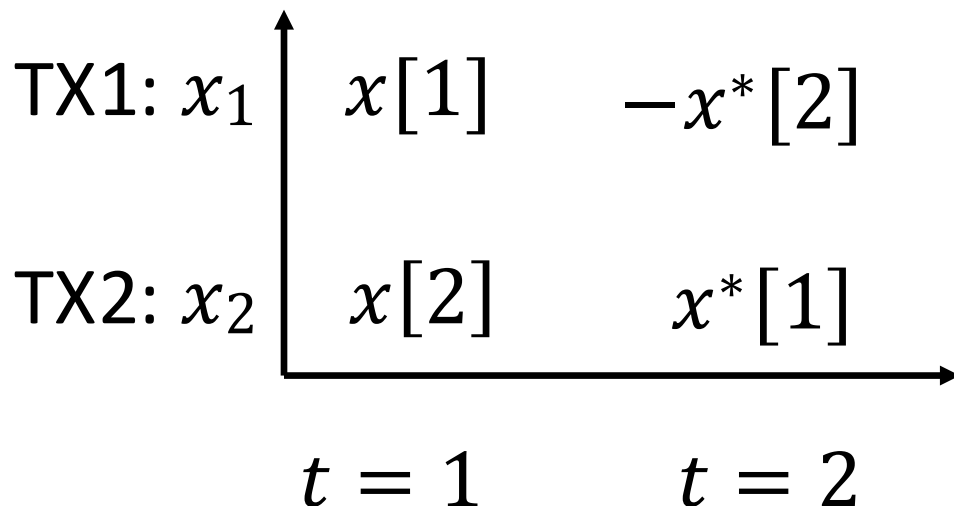
Transmitter Diversity - Normalize



What should we transmit on each antenna?

Solution: Use Space-Time Codes

Alamouti Codes:



$$h_1^*y[1] + h_2y^*[2] = (|h_1|^2 + |h_2|^2)x[1]$$

$$h_2^*y[1] - h_1y^*[2] = (|h_1|^2 + |h_2|^2)x[2]$$

Alamouti SNR

$$\begin{aligned}\text{SNR} &= \frac{\|\mathbf{h}\|^2}{\sigma_n^2} \text{E} \left\{ |x_1|^2 \right\} \\ &= \frac{\|\mathbf{h}^2\|}{\sigma_n^2} P_1,\end{aligned}$$

$$\frac{\|\mathbf{h}^2\|}{\sigma_n^2} P_1, \quad \frac{\|\mathbf{h}^2\|}{\sigma_n^2} P_2,$$

$$P_1 = P_2 = \frac{P}{2}.$$

$$\text{SNR} = \frac{P}{2} \frac{\|\mathbf{h}^2\|}{\sigma_n^2} = \frac{1}{2} \frac{\|\mathbf{h}^2\|}{\sigma_n^2} P$$

MIMO Gains

Multiplexing Gain:

- Send multiple packets at the same time
- $N \times N$ MIMO $\rightarrow \times N$ more packets
- Data Rate: $\propto N \log(SNR/N)$

Diversity Gain:

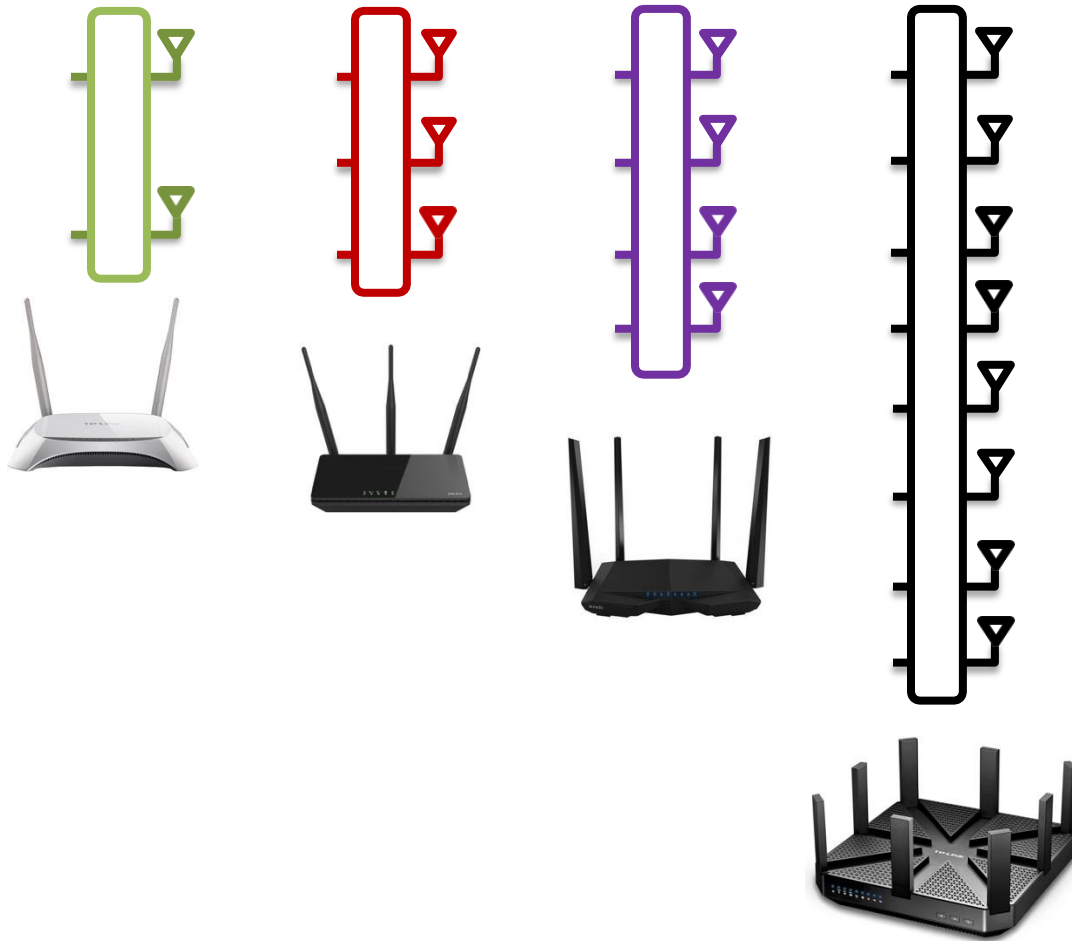
- Increase SNR of the received packets
- $N \times N$ MIMO $\rightarrow \times \log N^2$ data rate
- Data Rate $\propto \log(SNR \times N^2)$

Today's Class

- MIMO Channel
 - Channel representation
 - Preamble based channel estimation
- MIMO Gains
 - Receiver Diversity
 - Transmitter Diversity
 - Space-Time codes (Alamouti Codes)
- **MIMO Multiplexing Gains**
 - **Zero-forcing**
 - **SVD**

MIMO: Multiple TX-RX streams

WiFi Access Points 2 to 8 MIMO

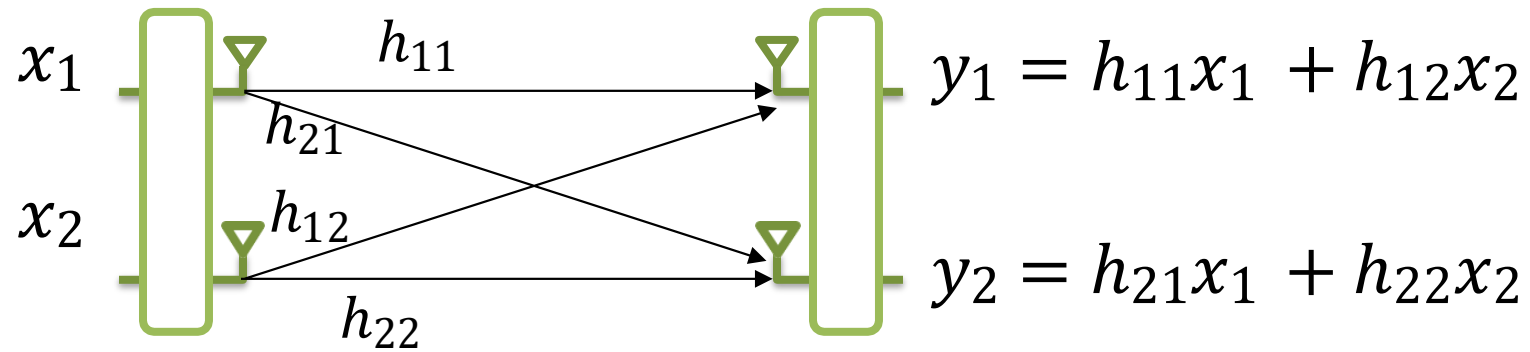


WiFi Devices 1 to 2 MIMO



- Power
- Form Factor & Antenna Separation
- Most mobile phones support 2 MIMO

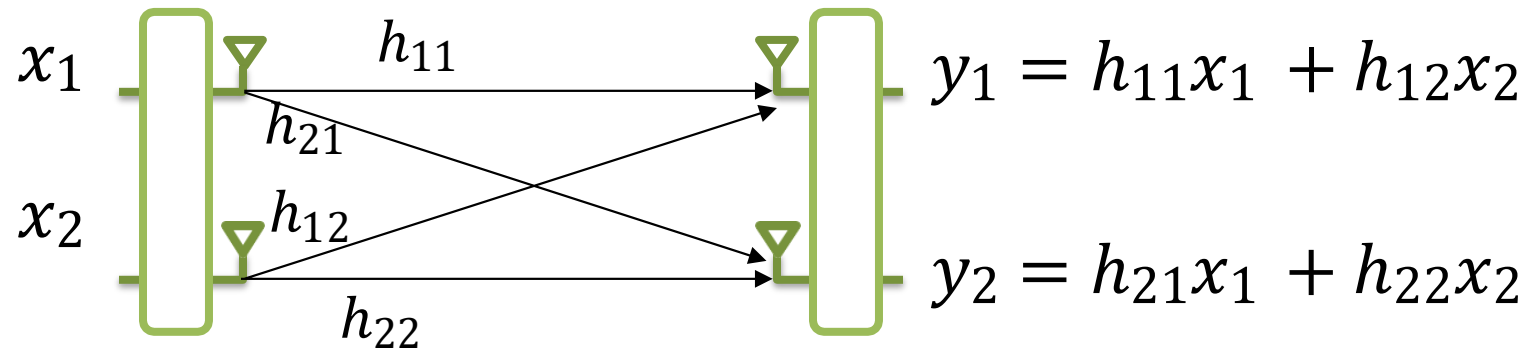
MIMO



$$\mathbf{y} = \mathbf{H}\mathbf{x} + \mathbf{n}$$

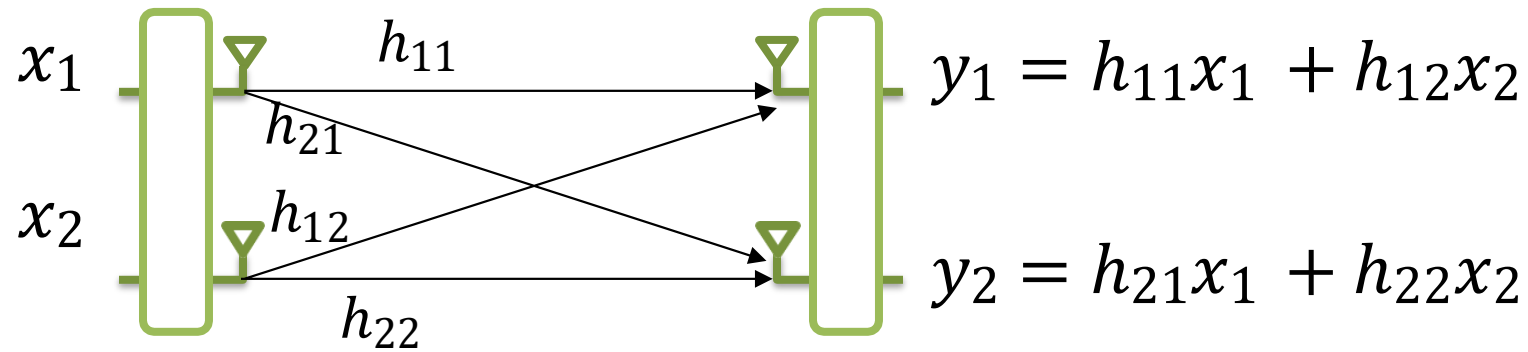
$$\tilde{\mathbf{x}} = \mathbf{H}^{-1}\mathbf{y} = \mathbf{H}^{-1}\mathbf{H}\mathbf{x} + \mathbf{H}^{-1}\mathbf{n} = \mathbf{x} + \mathbf{H}^{-1}\mathbf{n}$$

Zero-forcing based MIMO



$$\hat{\mathbf{x}} = (\mathbf{H}^H \mathbf{H})^{-1} \mathbf{H}^H \mathbf{y}$$

SVD-based MIMO



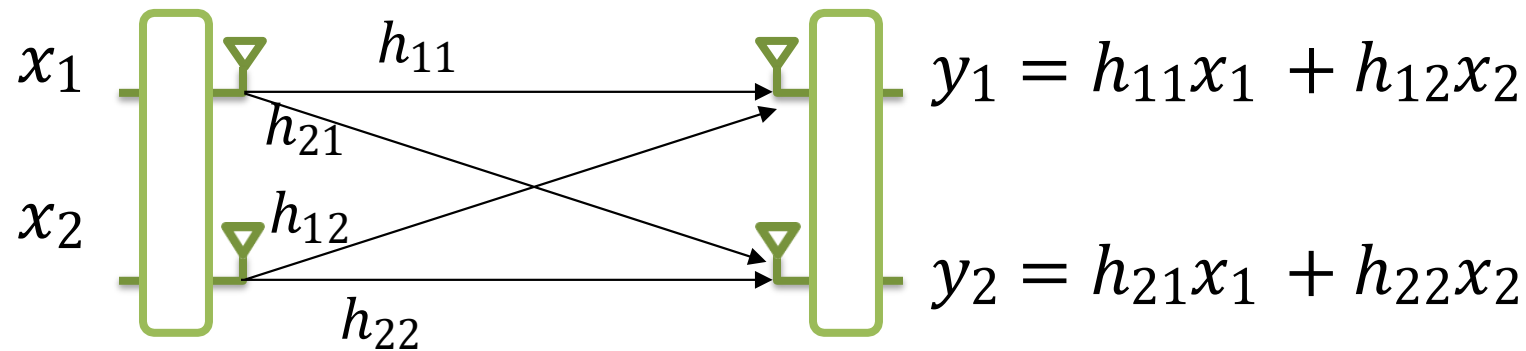
$$\mathbf{H} = \underbrace{\begin{bmatrix} u_{11} & u_{12} & \dots & u_{1t} \\ u_{21} & u_{22} & \dots & u_{2t} \\ \vdots & \vdots & \ddots & \vdots \\ \underbrace{u_{r1}}_{\mathbf{u}_1} & \underbrace{u_{r2}}_{\mathbf{u}_2} & \dots & \underbrace{u_{rt}}_{\mathbf{u}_t} \end{bmatrix}}_{\mathbf{U}} \underbrace{\begin{bmatrix} \sigma_1 & 0 & \dots & 0 \\ 0 & \sigma_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \sigma_t \end{bmatrix}}_{\mathbf{\Sigma}}$$

$$\underbrace{\begin{bmatrix} v_{11}^* & v_{21}^* & \dots & v_{t1}^* \\ v_{12}^* & v_{22}^* & \dots & v_{t2}^* \\ \vdots & \vdots & \ddots & \vdots \\ v_{1t}^* & v_{2t}^* & \dots & v_{tt}^* \end{bmatrix}}_{\mathbf{V}^H},$$

$\left. \begin{matrix} v_{11}^* & v_{21}^* & \dots & v_{t1}^* \end{matrix} \right\} \mathbf{v}_1^H$
 $\left. \begin{matrix} v_{12}^* & v_{22}^* & \dots & v_{t2}^* \end{matrix} \right\} \mathbf{v}_2^H$
 $\left. \begin{matrix} v_{1t}^* & v_{2t}^* & \dots & v_{tt}^* \end{matrix} \right\} \mathbf{v}_t^H$

$$= \mathbf{U}\mathbf{\Sigma}\mathbf{V}^H$$

SVD-based MIMO



$$\mathbf{H} = \underbrace{\begin{bmatrix} u_{11} & u_{12} & \dots & u_{1t} \\ u_{21} & u_{22} & \dots & u_{2t} \\ \vdots & \vdots & \ddots & \vdots \\ \underbrace{u_{r1}}_{\mathbf{u}_1} & \underbrace{u_{r2}}_{\mathbf{u}_2} & \dots & \underbrace{u_{rt}}_{\mathbf{u}_t} \end{bmatrix}}_{\mathbf{U}} \underbrace{\begin{bmatrix} \sigma_1 & 0 & \dots & 0 \\ 0 & \sigma_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \sigma_t \end{bmatrix}}_{\mathbf{\Sigma}}$$

$$\|\mathbf{u}_i\|^2 = \|\mathbf{v}_i\|^2 = 1, 1 \leq i \leq t$$

$$\mathbf{u}_i^H \mathbf{u}_j = \mathbf{v}_i^H \mathbf{v}_j = 0, i \neq j, 1 \leq i, j \leq t$$

$$\mathbf{V}^H \mathbf{V} = \mathbf{V} \mathbf{V}^H = \mathbf{I}_t$$

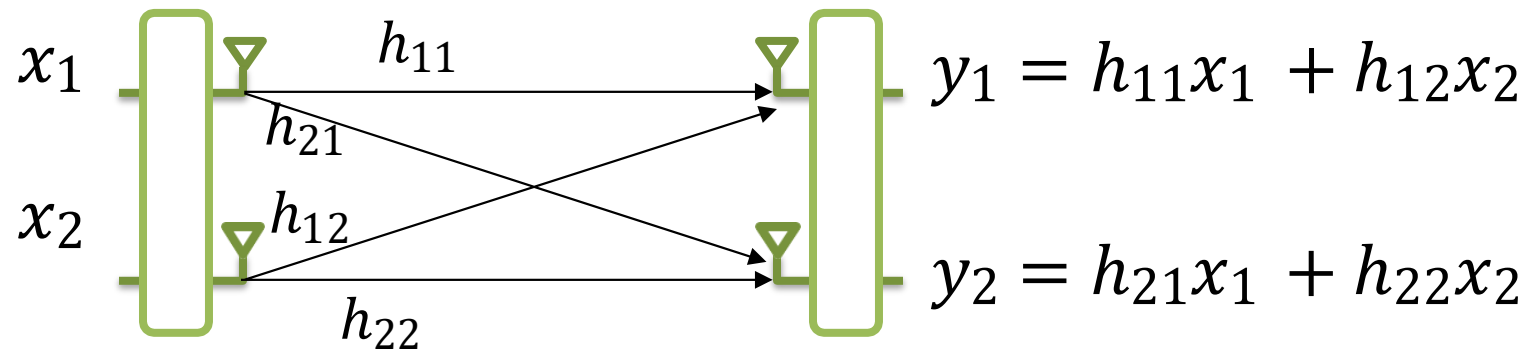
$$\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_t$$

$$\underbrace{\begin{bmatrix} v_{11}^* & v_{21}^* & \dots & v_{t1}^* \\ v_{12}^* & v_{22}^* & \dots & v_{t2}^* \\ \vdots & \vdots & \ddots & \vdots \\ v_{1t}^* & v_{2t}^* & \dots & v_{tt}^* \end{bmatrix}}_{\mathbf{V}^H},$$

$\left. \begin{array}{l} \vdots \\ \vdots \end{array} \right\} \mathbf{v}_1^H$
 $\left. \begin{array}{l} \vdots \\ \vdots \end{array} \right\} \mathbf{v}_2^H$
 $\left. \begin{array}{l} \vdots \\ \vdots \end{array} \right\} \mathbf{v}_t^H$

$$= \mathbf{U} \mathbf{\Sigma} \mathbf{V}^H$$

SVD-based MIMO



$$\mathbf{y} = \mathbf{H}\mathbf{x} + \mathbf{n}$$

$$\tilde{\mathbf{y}} = \mathbf{\Sigma}\mathbf{V}^H\mathbf{x} + \tilde{\mathbf{n}}$$

$$\mathbf{y} = \underbrace{\mathbf{U}\mathbf{\Sigma}\mathbf{V}^H}_{\mathbf{H}}\mathbf{x} + \mathbf{n}$$

$$= \mathbf{\Sigma}\mathbf{V}^H\mathbf{V}\tilde{\mathbf{x}} + \tilde{\mathbf{n}}$$

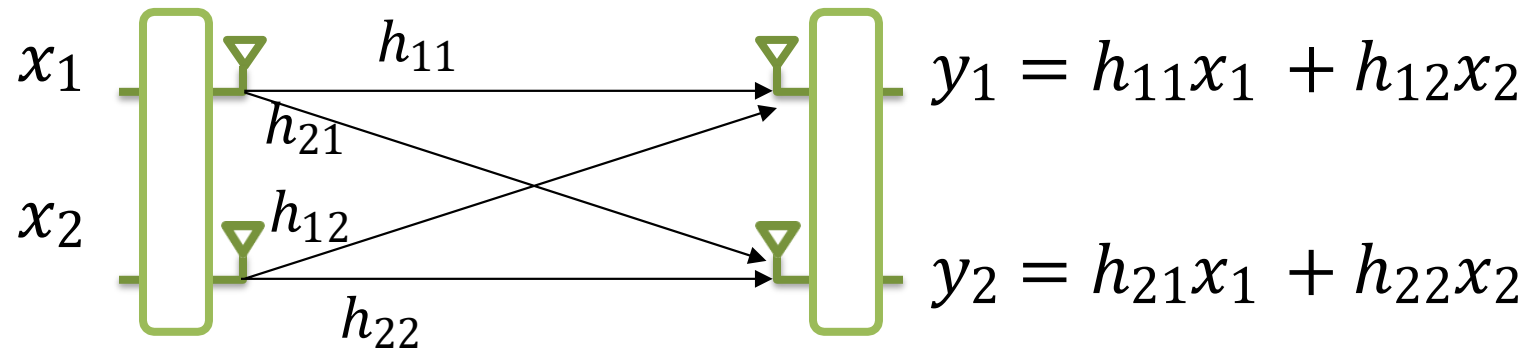
$$\underbrace{\mathbf{U}^H\mathbf{y}}_{\tilde{\mathbf{y}}} = \mathbf{U}^H (\mathbf{U}\mathbf{\Sigma}\mathbf{V}^H\mathbf{x} + \mathbf{n})$$

$$= \mathbf{\Sigma}\tilde{\mathbf{x}} + \tilde{\mathbf{n}}$$

$$\tilde{\mathbf{y}} = \underbrace{\mathbf{U}^H\mathbf{U}}_{\mathbf{I}_t}\mathbf{\Sigma}\mathbf{V}^H\mathbf{x} + \underbrace{\mathbf{U}^H\mathbf{n}}_{\tilde{\mathbf{n}}}$$

$$= \mathbf{\Sigma}\mathbf{V}^H\mathbf{x} + \tilde{\mathbf{n}}$$

SVD-based MIMO



$$\begin{bmatrix} \tilde{y}_1 \\ \tilde{y}_2 \\ \vdots \\ \tilde{y}_t \end{bmatrix} = \begin{bmatrix} \sigma_1 & 0 & \dots & 0 \\ 0 & \sigma_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \sigma_t \end{bmatrix} \begin{bmatrix} \tilde{x}_1 \\ \tilde{x}_2 \\ \vdots \\ \tilde{x}_t \end{bmatrix} + \begin{bmatrix} \tilde{n}_1 \\ \tilde{n}_2 \\ \vdots \\ \tilde{n}_t \end{bmatrix}$$

$$\tilde{y}_1 = \sigma_1 \tilde{x}_1 + \tilde{n}_1$$

$$\tilde{y}_2 = \sigma_2 \tilde{x}_2 + \tilde{n}_2$$

$$\vdots$$

$$\tilde{y}_t = \sigma_t \tilde{x}_t + \tilde{n}_t$$

$$C_i = \log_2 \left(1 + \frac{P_i \sigma_i^2}{\sigma_n^2} \right)$$

Total Capacity

$$C = \sum_{i=1}^t \log_2 \left(1 + \frac{P_i \sigma_i^2}{\sigma_i^2} \right)$$